

Andrei-Cătălin Popa

PHD STUDENT IN ASTROPHYSICS

Astrophysics Research Centre, Keele University, ST5 5BG, UK

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Education

Keele University

Keele, UK

PHD IN ASTROPHYSICS

Nov. 2026 – present

Supervisor: Dr. Pierre Maxted.

Thesis: *Calibrating Stellar Models with Chemical Tracers of Internal Mixing Processes.*

PLATO mission aims to discover and characterise rocky planets orbiting Sun-like stars, with stellar ages measured to 10% accuracy.

Developing a new method to calibrate next-generation stellar models using chemical tracers in binary star spectra to probe the efficiency of internal mixing processes (convective overshooting, atomic diffusion, and rotation) in Sun-like stars of known mass.

Université Côte d'Azur

Nice, France

MSC IN ASTROPHYSICS AND SPACE SCIENCE (ERASMUS MUNDUS JOINT MASTER'S PROGRAMME)

Mar. 2025 – Sept. 2026

Thesis: *Characterising M-dwarf Stars as Exoplanet Hosts in Preparation for the PLATO Mission.*

Advisors: Dr. Orlagh Creevey (OCA), Dr. Valentina D'Orazi (Tor Vergata), Dr. Ulrike Heiter (Uppsala).

Research Internship: *Fundamental Parameters of FGK Stars.*

Advisors: Dr. Orlagh Creevey (OCA) and Dr. Caroline Soubiran (Bordeaux).

Università degli Studi di Roma "Tor Vergata"

Rome, Italy

MSC IN ASTROPHYSICS AND SPACE SCIENCE (ERASMUS MUNDUS JOINT MASTER'S PROGRAMME)

Sept. 2024 – Mar. 2025

Consortium of four universities: Università Roma "Tor Vergata", Universität Bremen, Université Côte d'Azur, and University of Belgrade.

University College London

London, United Kingdom

MSCI IN PHYSICS (FIRST CLASS HONOURS)

Sept. 2018 – Jun. 2022

Supervisor: Prof. Daisuke Kawata (Mullard Space Science Laboratory).

Thesis: *Hunting for Intermediate-Mass Black Hole Binaries.*

Research Experience

Astrophysics Research Centre

Keele, United Kingdom

PHD RESEARCH: CALIBRATING STELLAR MODELS WITH CHEMICAL TRACERS OF INTERNAL MIXING PROCESSES

Nov. 2026 – present

- Supervised by Dr. Pierre Maxted. Research within the ESA PLATO mission consortium, focused on using chemical abundance tracers in binary stars to calibrate stellar structure and evolution models.
- Investigating internal mixing processes (convective overshooting, atomic diffusion, rotation) through precision constraints provided by binary star systems and PLATO lightcurves.

Observatoire de la Côte d'Azur

Nice, France

MASTER'S THESIS: CHARACTERISING M-DWARF STARS AS EXOPLANET HOSTS

Jan. 2026 – Sept. 2026

- Supervised by Dr. Orlagh Creevey (OCA), Dr. Ulrike Heiter (Uppsala University) and Dr. Valentina D'Orazi (Tor Vergata), within the Erasmus Mundus Joint Master's in Astrophysics and Space Science (MASS).
- Deriving effective temperatures of benchmark M-dwarf stars via spectral energy distribution (SED) fitting and bolometric flux calculations, using Gaia XP spectra and multi-band photometry.
- Building and testing Python pipelines for photometric and spectroscopic data comparison, integrating literature stellar parameters for cross-validation.
- Contributing to PLATO mission by refining fundamental parameters used in the mission's stellar characterisation framework.

Astrophysics Research Institute

Liverpool, United Kingdom

SUMMER RESEARCH PROJECT: WHITE DWARF COOLING SEQUENCES AND INFRARED COLOUR EXCESS

Jul. 2025 – Sept. 2025

- Supervised by Prof. Maurizio Salaris. Investigated whether the apparent broadening of the white dwarf cooling sequence in NGC 6397 reflects an intrinsic IR-excess population or statistical scatter.
- Analysed JWST and HST photometric data; constructed synthetic white dwarf populations from theoretical isochrones and a Salpeter IMF, with Monte Carlo uncertainty propagation.
- Developed Python routines to compare observed and simulated colour-magnitude diagrams, applying Gaussian Mixture Models (GMM) and Bayesian Information Criterion (BIC) analysis.

Scholarships & Honours

2026	STFC PhD Studentship , Science and Technology Facilities Council / UKRI	£21,805 p.a.
2024	Erasmus Mundus (EMJM) Scholarship , European Education and Culture Executive Agency	€33,600
2022	Summer Research Internship Scholarship , Dept. of Physics & Astronomy, UCL	£2,000
2022	"Teach Physics" Internship Award , Ogden Trust — Eastbury Community School, London	£1,400

Conference Presentations

Fundamental properties of M dwarfs

EAS Annual Meeting, Lausanne, CH

CONTRIBUTED TALK

July 2026

- Session SS25c: Optical Interferometry in the next decade
- *Authors:* A.C. Popa, O. Creevey, V. D'Orazi, U. Heiter

Teaching

Eastbury Community School

Barking, United Kingdom

SUMMER PHYSICS TEACHING INTERN — OGDEN TRUST "TEACH PHYSICS" PROGRAMME

Jun. – Jul. 2022

Taught A-level and GCSE physics through interactive sessions and lab exercises; created classroom materials aligned with UK physics curricula; supported Science Club activities.

Outreach

UCL Observatory

Mill Hill, United Kingdom

DEMONSTRATOR

Oct. 2019 – Sept. 2022

Trained to operate robotic and manual telescopes (14-inch Schmidt–Cassegrain; twin 24/18-inch refractors). Led public observing sessions for school groups and members of the public; developed science communication skills across diverse audiences.

Department of Physics & Astronomy, University College London

London, United Kingdom

OUTREACH INTERN

Jun. – Sept. 2022

Organised outreach activities in local schools and community groups; planned and delivered science communication events including UCL's public festival *Your Universe*; produced materials translating astrophysical research for non-specialist audiences.

Service & Committees

2025 **Student Representative**, Quality Evaluation Committee, Erasmus Mundus MASS Programme

Rome/Nice

Skills & Professional Memberships

Programming	Python, MATLAB, IDL
Astronomical Software	TOPCAT, SAO DS9, AstrolImageJ, VOSA, iSpec
Tools & Environments	Git, $\LaTeX$, UNIX/Linux
Memberships	Institute of Physics Royal Astronomical Society
	England Handball Referee Association Mensa Romania
Languages	English (Fluent) Romanian (Native) French (Intermediate)

References

Dr Pierre Maxted

Keele University

READER IN ASTROPHYSICS

contact: p.maxted AT keele.ac.uk

Dr Orlagh Creevey

Observatoire de la Côte d'Azur

RESEARCH ASTRONOMER

contact: orlagh.creevey AT oca.eu